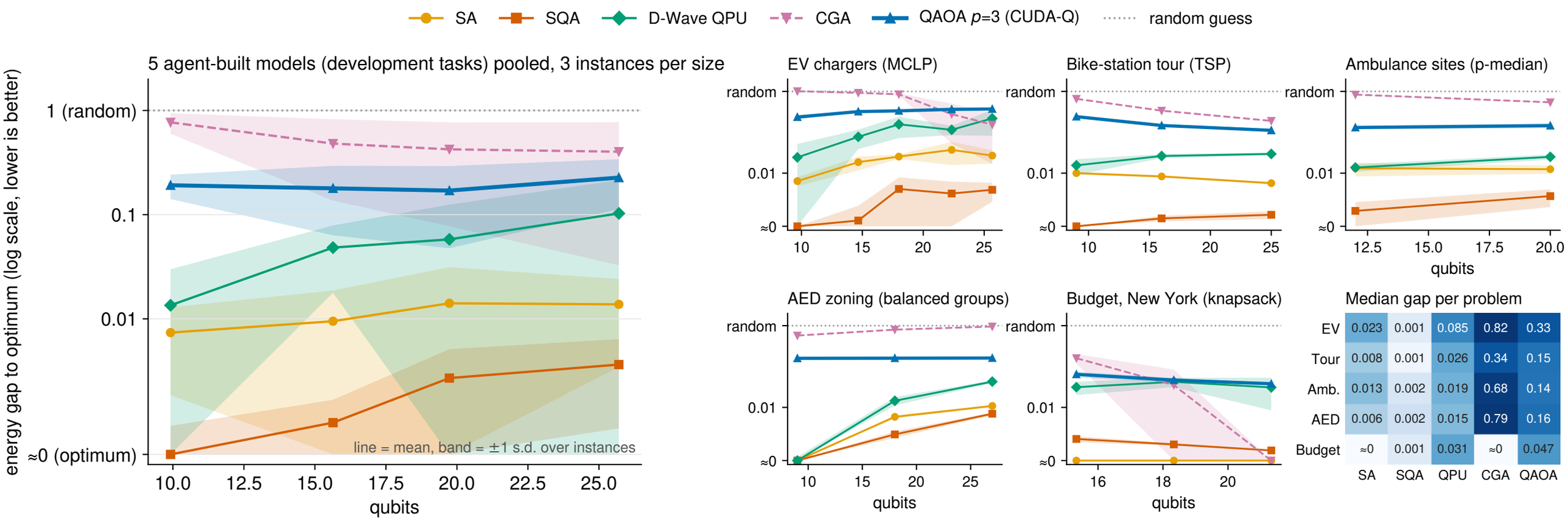
SA
SQA
D-Wave QPU
CGA
QAOA p=3 (CUDA-Q)
random guess
5 agent-built models (development tasks) pooled, 3 instances per size
energy gap to optimum (log scale, lower is better)
1 (random)
0.1
0.01
≈0 (optimum)
line = mean, band = ±1 s.d. over instances
10.0
12.5
15.0
17.5
20.0
22.5
25.0
qubits
EV chargers (MCLP)
Bike-station tour (TSP)
Ambulance sites (p-median)
AED zoning (balanced groups)
Budget, New York (knapsack)
random
0.01
≈0
10
15
20
25
12.5
15.0
17.5
20.0
16
18
qubits
Median gap per problem
EV 0.023 0.001 0.085 0.82 0.33
Tour 0.008 0.001 0.026 0.34 0.15
Amb. 0.013 0.002 0.019 0.68 0.14
AED 0.006 0.002 0.015 0.79 0.16
Budget ≈0 0.001 0.031 ≈0 0.047
SA
SQA
QPU
CGA
QAOA